

Direct-welded moment joint (no collar) - check sheet

Library dwj_lib.m, manual direct_weld_manual.md. Forces from
/home/winpatubuntu/scripting/structures/joint_calculations/etabs_joints/gg.txt, joint database built 25-Sep-2026 16:18:20; sheet
generated 25-Sep-2026 16:21:36. RSA combinations are excluded. Load combinations matching 'E[xy]' are reported as seismic,
the rest as gravity. One section per joint follows.

Summary

Joint	Column	Strong	Weak	Governing check	DCR	Case
10	50	[38 35]	[63 64]	F5s flange welds, K5 eff. length, strong	3.93	1.2D+L+1.6Lr
13	53	3	[14 16]	F5w flange welds, K5 eff. length, weak	3.26	1.2D+Ex+.3Ey+L+.2S Max

Joint 10

Joint 10, interior (X), 4 beams (h = 160, bf = 82) on HSS 200x100x4, column 50. Strong beams [38 35], weak beams [63 64]. Companion to dwj_lib.m and direct_weld_manual.md. Units N, mm throughout; results shown in kN and kNm. AISC 360-16, LRFD. Class (W N E S): M M M M; shear only on the shelf []; below the collar on a shear tab [].

Governing: F5s flange welds, K5 eff. length, strong, DCR 3.93, case 1.2D+L+1.6Lr. Gravity DCR = worst over the gravity combinations, seismic DCR = worst over combinations matching 'E[xy]'.

1. Index of limit states

Tag	Limit state	Reference	Capacity	Gravity DCR	Seismic DCR
B1	beam flange gross yielding	J4.1 (J4-1)	136.5 kN	0.78	0.73
F1s	flange eff. width yielding, strong	K2.2 plate local yield	31.2 kN	3.41	3.19
F5s	flange welds, K5 eff. length, strong	K5 + J2.4 (J2-3)	27.1 kN	3.93	3.67
F6s	face plastification, strong [INFO]	EN1993-1-8 T7.13	33.6 kN	3.17	2.96
F1w	flange eff. width yielding, weak	K2.2 plate local yield	15.6 kN	2.89	2.52
F5w	flange welds, K5 eff. length, weak	K5 + J2.4 (J2-3)	13.5 kN	3.33	2.91
F6w	face plastification, weak [INFO]	EN1993-1-8 T7.13	15.9 kN	2.84	2.48
C1s	column walls shear, strong	J4.2 (J4-3)	216.0 kN	0.02	0.24
C1w	column walls shear, weak	J4.2 (J4-3)	108.0 kN	0.02	0.31
W1	web fillet welds, 2 sides	J2.4 (J2-3)	177.4 kN	0.09	0.06
W2	wall vs web thickness, $t_w \leq F_u t / F_y$	K2.2 long. plate shear	6.40 mm	0.78	0.78
W3	beam web shear	G2.1 (G2-1)	120.0 kN	0.13	0.09
M1	beam flexure	F2.1 (F2-1)	27.878 kNm	0.58	0.54
M4	column flexure, strong	F7.1 (F7-1)	34.229 kNm	0.02	0.23
M5	column flexure, weak	F7.1 (F7-1)	21.089 kNm	0.01	0.24

DCR above 1.00 in red. Tags marked [MODEL] or [INFO] are engineering models, not code equations.

2. Data

Materials and sections

Item	Symbol	Value	Note
Elastic modulus	E	200000 MPa	
Electrode	FEXX	480 MPa	
Beam yield / tensile	Fy, Fu	250 / 400 MPa	
Beam depth	h	160 mm	
Beam flange width	bf	82 mm	
Beam flange thickness	tf	7.4 mm	
Beam web thickness	tw	5 mm	
Beam root radius	r	0 mm	
Beam plastic modulus	Zx	123900 mm ³	
Column side, strong direction	D	200 mm	HSS 200x100x4
Column side, weak direction	B	100 mm	
Column wall	t	4 mm	nominal
Column yield / tensile	Fy, Fu	250 / 400 MPa	
Column compression capacity	phiPn	not given	from ETABS; enables M6 (H1-1)

Welds and column faces

Item	Symbol	Value	How it is obtained
Flange weld type	fl_type	fillet	fillets both sides of each flange
Flange fillet leg	leg_fl	5 mm	each side
Web fillet leg	leg_web	4 mm	each side
Web weld length	Lw	145.2 mm	$h - 2 (tf + r)$, each side
Lever arm	z	152.6 mm	$h - tf$
Column outside corner radius	ro	6 mm	1.5 t when not known
Face width, strong beams	B_face	100 mm	column side the beam lands on
Face slenderness, strong	B/t	25.0	K2.2A: ≤ 35 flange, ≤ 40 web
Width ratio, strong	beta	0.820	bf / B_face
Side wall length, strong	H	200 mm	along the force
K5 effective weld length, strong	le	35.5 mm	$2 (10/(B/t)) (Fy t / Fyb tf) bf \leq 2 bf$
Face width, weak beams	B_face	200 mm	column side the beam lands on
Face slenderness, weak	B/t	50.0	K2.2A: ≤ 35 flange, ≤ 40 web
Width ratio, weak	beta	0.410	bf / B_face
Side wall length, weak	H	100 mm	along the force
K5 effective weld length, weak	le	17.7 mm	$2 (10/(B/t)) (Fy t / Fyb tf) bf \leq 2 bf$

AISC 360-16 Table K2.2A, limits of applicability

Limit	Value	
strong face, flange: $B/t \leq 35$	25.0	ok
strong face, web: $B/t \leq 40$	25.0	ok
strong face: $0.25 < \beta \leq 1.0$	0.82	ok
weak face, flange: $B/t \leq 35$	50.0	OUT
weak face, web: $B/t \leq 40$	50.0	OUT
weak face: $0.25 < \beta \leq 1.0$	0.41	ok
column $Fy \leq 360$ MPa	250	ok
column $Fy/Fu \leq 0.8$	0.62	ok

3. Demands

Quantity	How it is obtained	Gravity	Seismic
Load combination	from ETABS	1.2D+L+1.6Lr	1.2D+Ey+-.3Ex+L+.2S Max
Beam 38 moment (strong)	M3 at the beam end	16.22 kNm	8.45 kNm
Beam 35 moment (strong)	M3 at the beam end	15.21 kNm	15.16 kNm
Beam 63 moment (weak)	M3 at the beam end	6.88 kNm	2.28 kNm
Beam 64 moment (weak)	M3 at the beam end	6.61 kNm	3.88 kNm
Beam 38 axial	P at the beam end	-0.03 kN	0.41 kN
Beam 35 axial	P at the beam end	0.08 kN	-0.33 kN
Beam 63 axial	P at the beam end	-0.05 kN	0.28 kN
Beam 64 axial	P at the beam end	-0.02 kN	0.07 kN
Column-top moment, strong	M3 at the column top	0.54 kNm	6.78 kNm
Column-top moment, weak	other axis	0.25 kNm	1.55 kNm
Largest reaction	-F1 on the joint, column axes	15.37 kN	10.60 kN
Column axial	-P at the column top	41.63 kN	23.21 kN
Flange force, strong	M /z + N /2, demand of F1s	106.3 kN	99.5 kN
Flange force, weak	M /z + N /2, demand of F1w	45.1 kN	25.5 kN
Net force, strong	M_col / z, demand of C1s	3.5 kN	44.4 kN
Net force, weak	M_col / z, demand of C1w	1.7 kN	10.2 kN

The gravity and seismic columns are the combinations with the largest DCR of any check in each group.

4. Capacities, with numbers substituted

Tag	Reference	Substitution	Result
B1	J4.1 (J4-1)	$0.90 \times 250 \times 82 \times 7.4$	136.5 kN
F1s	K2.2 plate local yield	$0.95 \times \min(10/25 \times 250 \times 4 \times 82, 250 \times 7.4 \times 82)$	31.2 kN
F5s	K5 + J2.4 (J2-3)	$le=35.46: 0.75 \times 0.60 \times 480 \times 0.707 \times 5 \times 35.46$	27.1 kN
F6s	EN1993-1-8 T7.13	$kn=1.000: 1.000 \times 250 \times 4^2 \times (2 + 2.8 \times 0.820) / \sqrt{1 - 0.9 \times 0.820}$	33.6 kN
F1w	K2.2 plate local yield	$0.95 \times \min(10/50 \times 250 \times 4 \times 82, 250 \times 7.4 \times 82)$	15.6 kN
F5w	K5 + J2.4 (J2-3)	$le=17.73: 0.75 \times 0.60 \times 480 \times 0.707 \times 5 \times 17.73$	13.5 kN
F6w	EN1993-1-8 T7.13	$kn=1.000: 1.000 \times 250 \times 4^2 \times (2 + 2.8 \times 0.410) / \sqrt{1 - 0.9 \times 0.410}$	15.9 kN
C1s	J4.2 (J4-3)	$0.90 \times 0.60 \times 250 \times 2 \times 200 \times 4$	216.0 kN
C1w	J4.2 (J4-3)	$0.90 \times 0.60 \times 250 \times 2 \times 100 \times 4$	108.0 kN
W1	J2.4 (J2-3)	$0.75 \times 0.60 \times 480 \times 0.707 \times 4 \times 2 \times 145.2$	177.4 kN
W2	K2.2 long. plate shear	$400 \times 4 / 250$	6.40 mm
W3	G2.1 (G2-1)	$1.00 \times 0.60 \times 250 \times 160 \times 5$	120.0 kN
M1	F2.1 (F2-1)	$0.90 \times 123900 \times 250$	27.878 kNm
M4	F7.1 (F7-1)	$0.90 \times 152128 \times 250$	34.229 kNm
M5	F7.1 (F7-1)	$0.90 \times 93728 \times 250$	21.089 kNm

6. Checks to do first

1. F5s flange welds, K5 eff. length, strong (K5 + J2.4 (J2-3)): $l_e=35.46: 0.75 \times 0.60 \times 480 \times 0.707 \times 5 \times 35.46 = 27.1 \text{ kN}$, against a demand of 106.3 kN in 1.2D+L+1.6Lr.
2. B1 beam flange gross yielding (J4.1 (J4-1)): $0.90 \times 250 \times 82 \times 7.4 = 136.5 \text{ kN}$, against a demand of 106.3 kN in 1.2D+L+1.6Lr.
3. F1s flange eff. width yielding, strong (K2.2 plate local yield): $0.95 \times \min(10/25 \times 250 \times 4 \times 82, 250 \times 7.4 \times 82) = 31.2 \text{ kN}$, against a demand of 106.3 kN in 1.2D+L+1.6Lr.
4. W2 wall vs web thickness, $t_w \leq F_u t / F_y$ (K2.2 long. plate shear): $400 \times 4 / 250 = 6.40 \text{ mm}$, against a demand of 5.00 mm in 1.4D.

If these match, the units, the resistance factors and the forces are right, and the rest is the same arithmetic with different areas.

7. What is not on this sheet

Column P-M-M interaction unless ϕP_n is given (take it from ETABS), beam torsion, the web as a longitudinal plate under beam axial force, the design wall thickness $0.93 t$ of AISC B4.2 for ERW tubes (the nominal t is used), lateral-torsional buckling, fatigue, fire, and seismic detailing under AISC 341 (not prequalified under AISC 358). Faces outside Table K2.2A (section 2) are checked with the same equations for comparison only. F6 is an [INFO] cross-check from EN 1993-1-8, $\gamma_{M5} = 1.0$.

Joint 13

Joint 13, T, 3 beams ($h = 160$, $bf = 82$) on HSS 200x100x4, column 53. Strong beams 3, weak beams [14 16]. Companion to dwj_lib.m and direct_weld_manual.md. Units N, mm throughout; results shown in kN and kNm. AISC 360-16, LRFD. Class (W N E S): M M M NL; shear only on the shelf []; below the collar on a shear tab 70.

Governing: F5w flange welds, K5 eff. length, weak, DCR 3.26, case 1.2D+Ex+.3Ey+L+.2S Max. Gravity DCR = worst over the gravity combinations, seismic DCR = worst over combinations matching 'E[xy]'.

1. Index of limit states

Tag	Limit state	Reference	Capacity	Gravity DCR	Seismic DCR
B1	beam flange gross yielding	J4.1 (J4-1)	136.5 kN	0.24	0.50
F1s	flange eff. width yielding, strong	K2.2 plate local yield	31.2 kN	0.42	2.18
F5s	flange welds, K5 eff. length, strong	K5 + J2.4 (J2-3)	27.1 kN	0.48	2.51
F6s	face plastification, strong [INFO]	EN1993-1-8 T7.13	33.6 kN	0.39	2.02
F1w	flange eff. width yielding, weak	K2.2 plate local yield	15.6 kN	2.07	2.83
F5w	flange welds, K5 eff. length, weak	K5 + J2.4 (J2-3)	13.5 kN	2.38	3.26
F6w	face plastification, weak [INFO]	EN1993-1-8 T7.13	15.9 kN	2.03	2.78
C1s	column walls shear, strong	J4.2 (J4-3)	216.0 kN	0.07	0.31
C1w	column walls shear, weak	J4.2 (J4-3)	108.0 kN	0.03	0.26
W1	web fillet welds, 2 sides	J2.4 (J2-3)	177.4 kN	0.04	0.04
W2	wall vs web thickness, $t_w \leq F_u t / F_y$	K2.2 long. plate shear	6.40 mm	0.78	0.78
W3	beam web shear	G2.1 (G2-1)	120.0 kN	0.05	0.06
M1	beam flexure	F2.1 (F2-1)	27.878 kNm	0.18	0.37
M4	column flexure, strong	F7.1 (F7-1)	34.229 kNm	0.07	0.30
M5	column flexure, weak	F7.1 (F7-1)	21.089 kNm	0.03	0.21

DCR above 1.00 in red. Tags marked [MODEL] or [INFO] are engineering models, not code equations.

2. Data

Materials and sections

Item	Symbol	Value	Note
Elastic modulus	E	200000 MPa	
Electrode	FEXX	480 MPa	
Beam yield / tensile	Fy, Fu	250 / 400 MPa	
Beam depth	h	160 mm	
Beam flange width	bf	82 mm	
Beam flange thickness	tf	7.4 mm	
Beam web thickness	tw	5 mm	
Beam root radius	r	0 mm	
Beam plastic modulus	Zx	123900 mm ³	
Column side, strong direction	D	200 mm	HSS 200x100x4
Column side, weak direction	B	100 mm	
Column wall	t	4 mm	nominal
Column yield / tensile	Fy, Fu	250 / 400 MPa	
Column compression capacity	phiPn	not given	from ETABS; enables M6 (H1-1)

Welds and column faces

Item	Symbol	Value	How it is obtained
Flange weld type	fl_type	fillet	fillets both sides of each flange
Flange fillet leg	leg_fl	5 mm	each side
Web fillet leg	leg_web	4 mm	each side
Web weld length	Lw	145.2 mm	$h - 2 (tf + r)$, each side
Lever arm	z	152.6 mm	$h - tf$
Column outside corner radius	ro	6 mm	1.5 t when not known
Face width, strong beams	B_face	100 mm	column side the beam lands on
Face slenderness, strong	B/t	25.0	K2.2A: ≤ 35 flange, ≤ 40 web
Width ratio, strong	beta	0.820	bf / B_face
Side wall length, strong	H	200 mm	along the force
K5 effective weld length, strong	le	35.5 mm	$2 (10/(B/t)) (Fy t / Fyb tf) bf \leq 2 bf$
Face width, weak beams	B_face	200 mm	column side the beam lands on
Face slenderness, weak	B/t	50.0	K2.2A: ≤ 35 flange, ≤ 40 web
Width ratio, weak	beta	0.410	bf / B_face
Side wall length, weak	H	100 mm	along the force
K5 effective weld length, weak	le	17.7 mm	$2 (10/(B/t)) (Fy t / Fyb tf) bf \leq 2 bf$

AISC 360-16 Table K2.2A, limits of applicability

Limit	Value	
strong face, flange: $B/t \leq 35$	25.0	ok
strong face, web: $B/t \leq 40$	25.0	ok
strong face: $0.25 < \beta \leq 1.0$	0.82	ok
weak face, flange: $B/t \leq 35$	50.0	OUT
weak face, web: $B/t \leq 40$	50.0	OUT
weak face: $0.25 < \beta \leq 1.0$	0.41	ok
column $Fy \leq 360$ MPa	250	ok
column $Fy/Fu \leq 0.8$	0.62	ok

3. Demands

Quantity	How it is obtained	Gravity	Seismic
Load combination	from ETABS	1.2D+L+1.6Lr	1.2D+Ex+.3Ey+L+.2S Max
Beam 3 moment (strong)	M3 at the beam end	1.98 kNm	1.69 kNm
Beam 14 moment (weak)	M3 at the beam end	4.91 kNm	6.71 kNm
Beam 16 moment (weak)	M3 at the beam end	4.44 kNm	2.28 kNm
Beam 3 axial	P at the beam end	-0.05 kN	0.01 kN
Beam 14 axial	P at the beam end	-0.13 kN	-0.18 kN
Beam 16 axial	P at the beam end	-0.05 kN	0.20 kN
Column-top moment, strong	M3 at the column top	2.25 kNm	1.55 kNm
Column-top moment, weak	other axis	0.57 kNm	4.32 kNm
Largest reaction	-F1 on the joint, column axes	6.25 kN	4.85 kN
Column axial	-P at the column top	16.68 kN	8.50 kN
Flange force, strong	M /z + N /2, demand of F1s	13.0 kN	11.1 kN
Flange force, weak	M /z + N /2, demand of F1w	32.3 kN	44.1 kN
Net force, strong	M_col / z, demand of C1s	14.7 kN	10.1 kN
Net force, weak	M_col / z, demand of C1w	3.7 kN	28.3 kN

The gravity and seismic columns are the combinations with the largest DCR of any check in each group.

4. Capacities, with numbers substituted

Tag	Reference	Substitution	Result
B1	J4.1 (J4-1)	$0.90 \times 250 \times 82 \times 7.4$	136.5 kN
F1s	K2.2 plate local yield	$0.95 \times \min(10/25 \times 250 \times 4 \times 82, 250 \times 7.4 \times 82)$	31.2 kN
F5s	K5 + J2.4 (J2-3)	$le=35.46: 0.75 \times 0.60 \times 480 \times 0.707 \times 5 \times 35.46$	27.1 kN
F6s	EN1993-1-8 T7.13	$kn=1.000: 1.000 \times 250 \times 4^2 \times (2 + 2.8 \times 0.820) / \sqrt{1 - 0.9 \times 0.820}$	33.6 kN
F1w	K2.2 plate local yield	$0.95 \times \min(10/50 \times 250 \times 4 \times 82, 250 \times 7.4 \times 82)$	15.6 kN
F5w	K5 + J2.4 (J2-3)	$le=17.73: 0.75 \times 0.60 \times 480 \times 0.707 \times 5 \times 17.73$	13.5 kN
F6w	EN1993-1-8 T7.13	$kn=1.000: 1.000 \times 250 \times 4^2 \times (2 + 2.8 \times 0.410) / \sqrt{1 - 0.9 \times 0.410}$	15.9 kN
C1s	J4.2 (J4-3)	$0.90 \times 0.60 \times 250 \times 2 \times 200 \times 4$	216.0 kN
C1w	J4.2 (J4-3)	$0.90 \times 0.60 \times 250 \times 2 \times 100 \times 4$	108.0 kN
W1	J2.4 (J2-3)	$0.75 \times 0.60 \times 480 \times 0.707 \times 4 \times 2 \times 145.2$	177.4 kN
W2	K2.2 long. plate shear	$400 \times 4 / 250$	6.40 mm
W3	G2.1 (G2-1)	$1.00 \times 0.60 \times 250 \times 160 \times 5$	120.0 kN
M1	F2.1 (F2-1)	$0.90 \times 123900 \times 250$	27.878 kNm
M4	F7.1 (F7-1)	$0.90 \times 152128 \times 250$	34.229 kNm
M5	F7.1 (F7-1)	$0.90 \times 93728 \times 250$	21.089 kNm

6. Checks to do first

1. F5w flange welds, K5 eff. length, weak (K5 + J2.4 (J2-3)): $l_e=17.73: 0.75 \times 0.60 \times 480 \times 0.707 \times 5 \times 17.73 = 13.5 \text{ kN}$, against a demand of 44.1 kN in $1.2D+Ex+.3Ey+L+.2S$ Max.
2. B1 beam flange gross yielding (J4.1 (J4-1)): $0.90 \times 250 \times 82 \times 7.4 = 136.5 \text{ kN}$, against a demand of 67.9 kN in $1.3D+-Ey+.3Ex+L+.2S$ Max.
3. F1s flange eff. width yielding, strong (K2.2 plate local yield): $0.95 \times \min(10/25 \times 250 \times 4 \times 82, 250 \times 7.4 \times 82) = 31.2 \text{ kN}$, against a demand of 67.9 kN in $1.3D+-Ey+.3Ex+L+.2S$ Max.
4. W2 wall vs web thickness, $t_w \leq F_u t/F_y$ (K2.2 long. plate shear): $400 \times 4 / 250 = 6.40 \text{ mm}$, against a demand of 5.00 mm in 1.4D.

If these match, the units, the resistance factors and the forces are right, and the rest is the same arithmetic with different areas.

7. What is not on this sheet

Column P-M-M interaction unless ϕP_n is given (take it from ETABS), beam torsion, the web as a longitudinal plate under beam axial force, the design wall thickness $0.93 t$ of AISC B4.2 for ERW tubes (the nominal t is used), lateral-torsional buckling, fatigue, fire, and seismic detailing under AISC 341 (not prequalified under AISC 358). Faces outside Table K2.2A (section 2) are checked with the same equations for comparison only. F6 is an [INFO] cross-check from EN 1993-1-8, $\gamma_{M5} = 1.0$.